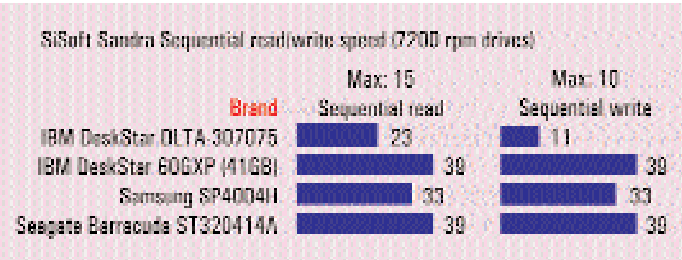
rates in sequential operations are faster than that of random operations as the drive does not have to reposition the read-write head each time it accesses a new file. Also, the speeds while reading are faster than that obtained when writing to the drive. In this test, the best performer was once again the **40 GB IBM Deskstar 60 GXP** hard disk, which scored a blazing 39



MBps in sequential read and 7 MBps in the random read test. When writing, this score dropped to 11 MBps and 9 MBps during the sequential and random write test, respectively. These are very good scores and indicate a usage of about 40 per cent of the entire bandwidth available in the ATA/100 interface. Though this seems rather low, this is the practical maximum that a single 7200-rpm hard disk can achieve. When evaluating the 5400-rpm drives, the under-utilisation of this bandwidth is far more pronounced due to the lower spindle speeds.

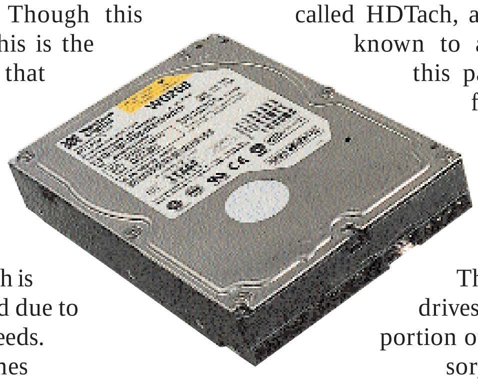
The access times logged by these 7200-rpm drives were between 5 ms and 9 ms and though this isn't much of a variation numerically, it does translate into the drive taking longer to access a file located on the hard disk. Therefore, in case of the 20 GB Seagate Barracuda, the access time of 5 ms means that it is ideally suited for applications where many files are accessed on the drives—applications such as multimedia authoring, databases, etc. In real world terms, a faster access time means quicker response of the hard disk, which is evident when applications start up faster and are more responsive when tasks are run.

What other factors affect performance?

There are other factors that are not directly related to the data transfer speed and access times of the hard disk but which affect the overall performance of applications running on your hard disk. One such parameter is the CPU utilisation of the hard disk.

All hard disks manufactured today use a technology called DMA (Direct Memory Access) which is a mode of data transfer where the hard disk orchestrates all data flow between itself and the rest of the system—the main CPU of your system plays little or no part in this. Therefore the level to which the hard disk depends on the main processor during data transfers would affect the performance of the rest of the system when the hard disk is being accessed. This is known as CPU utilisation and is expressed as a percentage.

The CPU utilisation of the drives was measured by a benchmarking program called HDTach, a program which is known to accurately measure this parameter. Here, we found that these 7,200-rpm drives posted a CPU utilisation ranging between 3.6 to 4.9 per cent. This indicates that the drives used a very small portion of the system processor, which, in turn, implies that the rest of your system runs smoothly even while the hard disk is being accessed, as the hard disk uses a very small portion of the main processor's power.



Western Digital WA Caviar WD200: this 5400-rpm drive carries an amply large 2 MB buffer onboard

Another interesting point seen in the results was that the CPU utilisation posted by SiSoft Sandra was significantly different from those logged by HDTach. This is because different applications write different type of data to the hard disk, which accounts for the variation in the scores logged by the two programs.

VALUE DRIVES (5400-RPM)

Over to the 5400-rpm category, these are the hard disks with wide variation, both, in terms of price as well as capacity. These are generally targeted at the value market where not much stress is placed on data transfer speeds, but on the amount of storage space available for a given price. Let's take a look at the drives that proved to be worthy performers here.

How a Hard Disk Works

Hard disks are used as a repository in which vast amounts of information can be stored. Unlike RAM, this information is permanent in that it is retained even after the computer is powered down. This is done using the principles of magnetic storage, which is similar to that used in retrieving and storing information on conventional tapes. Hard disks allow data to be stored at far denser levels and can be accessed very quickly. The working of a hard disk can be understood easily by drawing a parallel to the conventional audio or videotape. In an audiotape, a special magnetic material is coated onto a thin plastic strip whereas in a hard disk, the magnetic material is layered onto an aluminium or glass platter, which is polished to mirror smoothness. In an audiotape, the reading and recording head actually touches the surface directly during the read-write process but in hard disks, the head actually 'flies' microns above the surface of the platter and is never really allowed to touch the surface of the hard disk. Just fathom the speeds at which hard disks operate: the tape in a cassette recorder moves at a speed of about 2 inches per second while the platters of the hard disk can spin under the head at rates of up to 3,000 inches per second, translating into a speed of 225 kmph!

Application performance

Here, too, we ran high-end disk Winmark 99 as the primary benchmarking tool for evaluating the performance of the drive in applications such as multimedia authoring and software that especially stress the